
***Using Fourier Analysis for frequency
decomposition and harmonic comparison***

Mathematics Analysis and Approaches (HL)

Exploration

Number of pages: 20

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Introduction

Once while learning to play a song on my guitar through an online tutorial, I noticed that whenever I plucked the low E string, there was a faint vibration in the high E string too. Initially, I attributed this to resonance. However, I wondered how 2 string so different in characteristics like density and thickness, produce the same frequency. Knowing little about harmonics and music theory, I wanted to know how this occurred. So, I researched and found out about harmonics and their relation to frequencies of vibrating objects. I also learned about a mathematical approach to understanding these harmonic relationships called Fourier analysis. This link between math and music fascinated me. Since we learned about waves and functions in Mathematics AA HL, I decided to take this newfound knowledge one step further and explore Fourier analysis for my IA.

Aim:

The primary aim of this research is **to investigate the harmonic relationships of the high E and low E strings of a guitar using Fourier series analysis.**

Nomenclature:

Notation	Meaning
$f(x)$	Modelled function of high E string
$F(x)$	Fourier series of $f(x)$
A_0	Vertical shift above/below X-axis
k	Index of summation
n	Number of the harmonic
A_n	Amplitude or strength of the cosine waves that make up the function
B_n	Amplitude or strength of the sine waves that make up the function
L	Total time interval of audio sample
$g(x)$	Modelled function of low E string

Table 1: Nomenclature of notations used

Background:

Sound and Harmonics

When plucking a guitar string, it will not always vibrate in its entirety. It may vibrate in pieces of the string, and so it creates a series of higher frequencies known as **harmonics** (Lotha). The first harmonic of a guitar string is when it vibrates along its entire length, called fundamental frequency. The second harmonic is the vibration of the string along the length half of its fundamental frequency vibration and, therefore has a twice the frequency vibration. 2 consecutive harmonics are said to be an octave apart, meaning that one vibrates at twice the frequency of the other. So, the fundamental and second harmonic are 1 octave apart. The third harmonic vibrates in thirds, producing a pitch that is one and a half octaves higher than the fundamental (Hein). These harmonics are always in whole number ratios to each other. They overlap with one another, creating a rich and complex sound. Comparing two strings, like the low E and high E strings in this exploration, their fundamental frequencies and harmonics are related in specific ways. The two strings are two octaves apart, which means the frequency of the high E string is four times that of the low E string, which is correct since the fundamental frequency of the low E string is 82.407 Hz and of the high E string is 329.63 Hz as per the internationally accepted Scientific Pitch Notation (SPN) (Liu et al.). This is also what I expect the values of my guitar string frequency to be. Their harmonics resonate with each other, giving rise to the phenomenon of harmonic interaction. Such a relationship can be proved and analysed with mathematical tools such as Fourier analysis, which I have done in this exploration.

Fourier analysis:

I researched the Fourier analysis method introduced by Joseph Fourier in ‘The Analytical Theory of Heat’ (1822) (Schreiber), including its two types: Fourier series and Fourier transform. I found out that to choose the correct type of analysis, I had to see whether the wave was a continuous time signal or discrete time signal (“Cross-Rate Operations”).

Audacity gave a discrete time signal, but the step length was almost negligible, 0.000004 s, and I needed to take a large part of the audio for processing as compared to the step length. Hence, I assumed the wave to be continuous, and since I chose the part that had almost no damping, I safely assumed it to be periodic too (Browdues). Since both the requirements were met, I chose the Fourier series analysis.

Essentially what the Fourier series does is breaking down a complex waveform, like the sound wave of a vibrating guitar string, into simpler sine and cosine waves, with specific frequencies, amplitudes and phases (Cheever, Hayes). By analysing these components, I can find out which frequencies are present in the wave, and thus obtain information about its harmonic structure via frequency decomposition. The general formula for Fourier series $F(x)$ of a periodic function $f(x)$ with time period 2π , within limits $(-\pi, \pi)$ is:

$$F(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nx + \sum_{n=1}^{\infty} B_n \sin nx$$

Where:

$$A_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$A_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$B_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Here, A_0 is the average level or ‘baseline’ of the signal, which is basically shows the vertical shift from the X-axis. ‘ n ’ is the number of harmonics, which can go from fundamental where $n=1$ to infinity. A_n and B_n are the amplitude of the cosine and sine waves of the n^{th} harmonic frequency making up the function respectively. They basically show the strength of each harmonic that adds up to form the wave.

For my exploration, to avoid extrapolating into negative and for ease of calculation, I have made a few adjustments to account for the fact that the $f(x)$ has a period equal to whole time interval of audio, L (Lemmaxiom):

$$F(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{2\pi nx}{L} + \sum_{n=1}^{\infty} B_n \sin \frac{2\pi nx}{L} \quad \text{Eq. 1}$$

And since for Fourier decomposition over the interval $(0, L)$, the signal is treated as if L is the effective period of the function, which would change the coefficients to:

$$A_0 = \frac{1}{L} \int_0^L f(x) dx \quad \text{Eq. 2}$$

$$A_n = \frac{1}{L} \int_0^L f(x) \cos \frac{2\pi nx}{L} dx \quad \text{Eq. 3}$$

$$B_n = \frac{1}{L} \int_0^L f(x) \sin \frac{2\pi nx}{L} dx \quad \text{Eq. 4}$$

In this investigation, in short, I will take a recorded audio sample, make a function resembling that sample, perform Fourier series analysis on that sample and get the dominant frequencies. I will do this twice, one on the high E string and one on the low E string, and compared their harmonic structure so see how much part of the sound in each has a common harmonic. I chose these both since these are the same note, which assures some commonality in harmonic structure for me to investigate, and also as these were the strings that initially resonated to inspire this IA. As an added measure of accuracy, I will compare those frequencies with literature values and also with frequencies given by the audio recording software that performs the same Fourier analysis. I expect that the frequencies of both the strings that I will find through my exploration will be in a whole number ratio to each other, and the presence of the fundamental frequency of the high E string in the sound of the low E string provides an answer for the resonance of the high E string when I played the low E string.

Procedure:

High E string: Calculations and Analysis

I began by recording myself pluck the high E string and low E string on Audacity (Audacity Team, 2024) software. This gave me the sound wave plotted in a graph, with X-axis as time and Y-axis as relative amplitude ranging between -1 and +1 (shown in Figure 1).

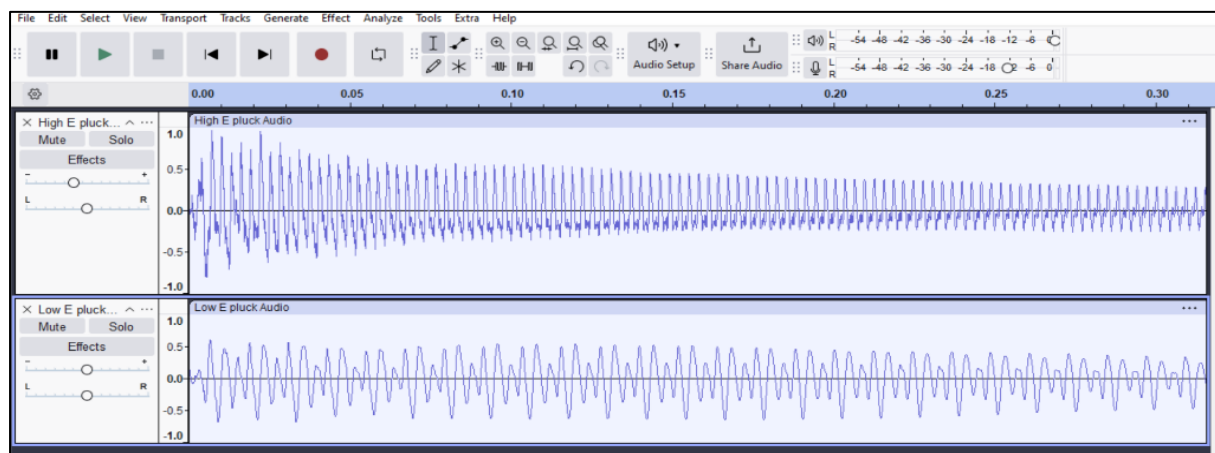


Figure 1: Waveform of both waves recorded on Audacity

Then, I observed shorter parts of the curves' carefully, noting amplitude, symmetry and the visual graphics, and noticed that they loosely resembled a negative ramp sawtooth wave. Hence, I decided to use the general function of a negative ramp sawtooth wave to deduce a function that approximates the sound wave as close as possible. Since there was no exact way to express the sound waves precisely as mathematical equations, I selected a part that looked as much as a periodic, non-damped function as possible, then, in Desmos' graphing calculator (Desmos), I worked out a function using the general function of negative ramp sawtooth wave ("A Sawtooth Wave") and trial and error method and tried modelling the curve to resemble the chosen part, which can be seen to be a good fit in Figure 2 below.

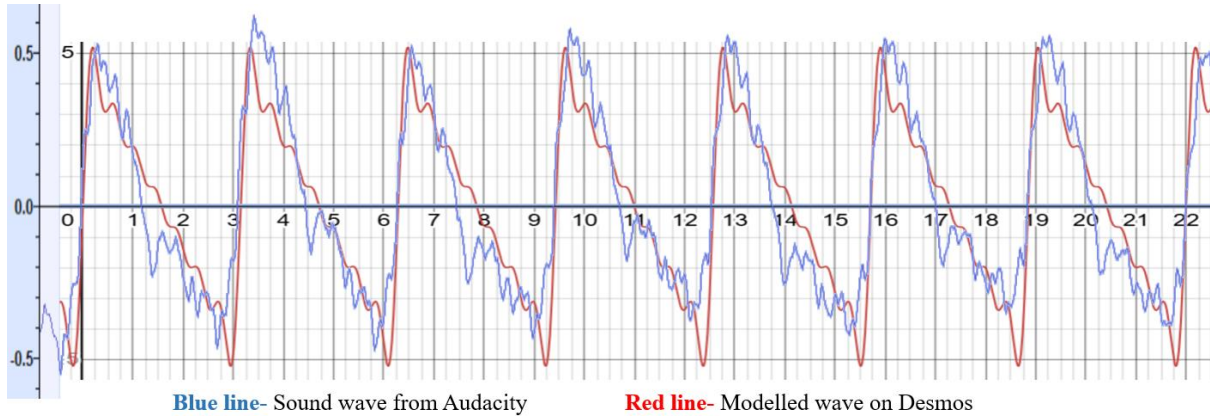


Figure 2: Matching the modelled wave to the original wave

The following is the model's equation $f(x)$:

$$f(x) = 0.315 \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) \quad \text{Eq. 5}$$

In Figure 2, I took 7 complete oscillations to analyse, since it gives Audacity enough data to analyse and Fourier analysis can be automatically computed by it, which will help check the accuracy of my result later on in the exploration. The interval of this audio sample was from 0.052 s to 0.074 s; hence L (in Eq. 2 to Eq. 4) = **0.022 s**, but to simplify modelling the equations and calculations, 22 units was taken.

To be completely sure of periodicity and to prevent any damping of the wave form affecting the analysis, I used a technique called windowing (Braun). Windowing is the technique to reduce errors in frequency analysis by looking at only part of the signal. Using this, I could 'force' periodicity to the wave, since it essentially smoothens the ends of the wave between the desired limits and this sample can now be repeated, thus forming a periodic wave ("Siemens Digital Industries Software Community"). Otherwise, spectral leakage may occur, wherein an energy extracted from one frequency "leaks" into other frequencies, distorting the frequency spectrum. This also causes frequency smearing, where sharp peaks in the frequency actually "spread out"(Mario). I selected the Hann window for its ability to minimize both

spectral leakage and frequency smearing and thereby give the results a good balance between accuracy and clarity. The effect can be visualized as follows:

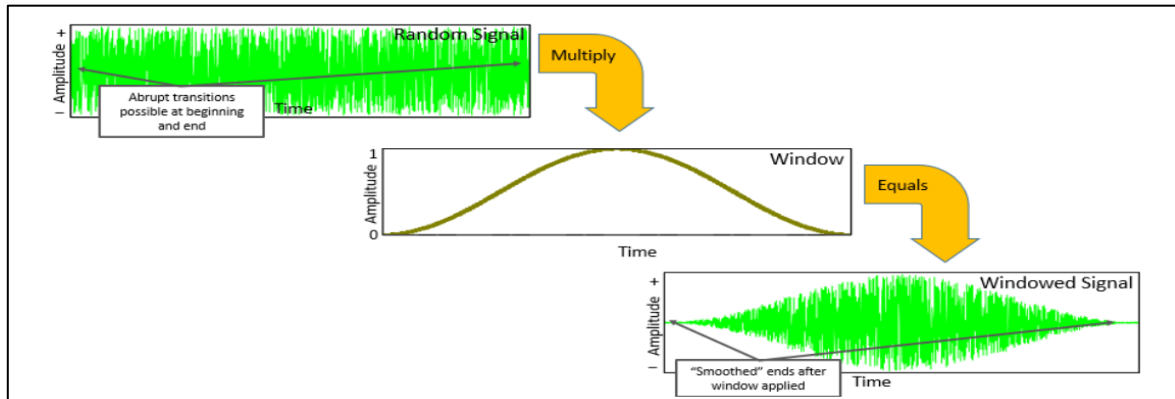


Figure 3: How the Hann window works (“Siemens Digital Industries Software Community”)

The Hann windowed function was:

$$f_{Hann}(x) = f(x) \times w(x)$$

Where the windowing function $w(x) = 0.5 \times (1 - \cos \frac{2\pi x}{L})$, making the function:

$$f_{Hann}(x) = 0.315 \left(\frac{\sin 2x}{2} + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) \times 0.5 \times \left(1 - \cos \frac{2\pi x}{L} \right)$$

This can be visually compared to the original function as:

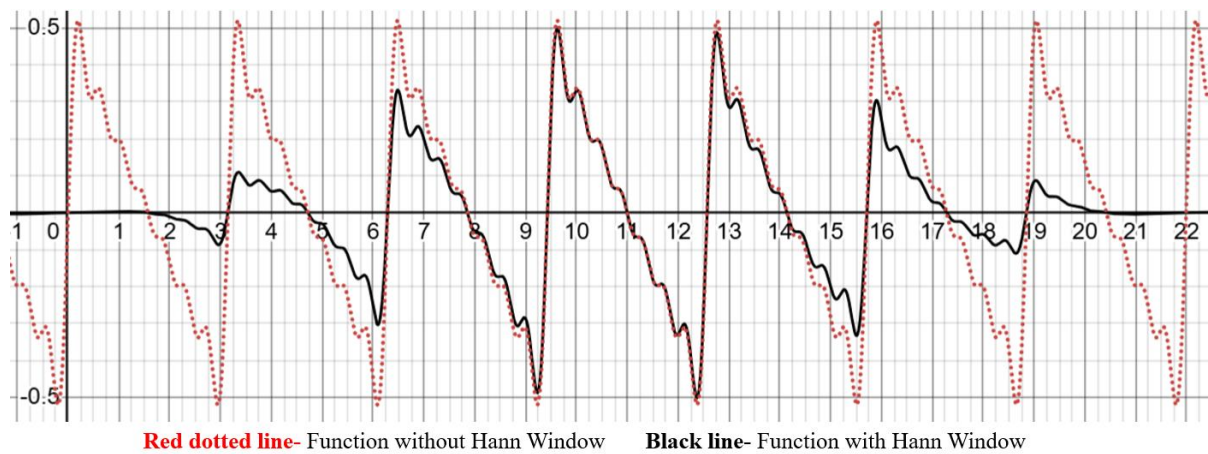


Figure 4: Hann windowed function visualised

Next, to calculate coefficients, first for A_0 , I solved the integral to get an answer in general terms as follows:

$$A_0 = \frac{1}{L} \int_0^L f(x) \times w(x) dx$$

$$A_0 = \frac{1}{L} \int_0^L a \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) \times \left(1 - \cos \frac{2\pi x}{L} \right) dx$$

$$A_0 = \frac{a}{L} \int_0^L \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) - \left(\cos \frac{2\pi x}{L} \right) \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) dx$$

Splitting the integrals into 2, I_1 and I_2 like:

$$A_0 = (I_1 - I_2)$$

Generalising I_1 in terms of k , where $k \in \mathbb{Z}^+$:

$$\begin{aligned} &= \frac{a}{L} \int_0^L \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) dx \\ &= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \sin 2kx dx \end{aligned} \quad k \in \mathbb{Z}^+$$

Now integrating in terms of k :

$$\begin{aligned} &= \frac{a}{L \cdot k} \left(-\frac{1}{2k} \cos 2kx \Big|_0^L \right) \\ &= \frac{a}{L \cdot k} \left(-\frac{1}{2k} \cos 2kL - \frac{1}{2k} \cos 0 \right) \\ &= \frac{a}{L \cdot k} \times \left(-\frac{1}{2k} \right) (1 - \cos 2kL) \end{aligned}$$

Thus, I_1 becomes:

$$I_1 = \sum_{k=1}^7 \frac{a}{L \cdot k} \times \left(-\frac{1}{2k} \right) (1 - \cos 2kL) \quad k \in \mathbb{Z}^+ \quad \underline{\text{Eq. 6}}$$

Where:

a = Amplitude of the wave being analysed

L = Time interval of sampled wave

n = Number of desired harmonics

Now, generally solving I_2 :

$$I_2 = -\frac{a}{L} \int_0^L \left(\cos \frac{2\pi x}{L} \right) \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right)$$

Which can be generally written as:

$$I_2 = -\frac{a}{L} \int_0^L \left(\cos \frac{2\pi x}{L} \right) \times \left(\frac{1}{k} \sum_{k=1}^7 \sin 2kx \right) dx$$

Now, since there are trigonometric terms being multiplied, I chose to use the following identity:

$$\sin A \cos B = \frac{1}{2} \sin (A + B) + \sin (A - B)$$

Substituting in this $A = kx$ and $B = \frac{2\pi x}{L}$ for each k , I wrote the general integrand for each k as:

$$\sin kx \times \cos \frac{2\pi x}{L} = \frac{1}{2} \sin \left(x \left(k + \frac{2\pi}{L} \right) \right) + \sin \left(x \left(k - \frac{2\pi}{L} \right) \right)$$

Now, substituting this in the integral:

$$\begin{aligned} I_2 &= -\frac{a}{L} \int_0^L \frac{1}{2} \sin \left(x \left(k + \frac{2\pi}{L} \right) \right) + \sin \left(x \left(k - \frac{2\pi}{L} \right) \right) dx \\ &= -\frac{a}{L} \int_0^L \frac{1}{2} \sin \left(x \left(k + \frac{2\pi}{L} \right) \right) dx - \frac{a}{L} \int_0^L \sin \left(x \left(k - \frac{2\pi}{L} \right) \right) dx \\ &= \left(-\frac{a}{L} \times -\frac{1}{k + \frac{2\pi}{L}} \left(\cos \left(x \left(k + \frac{2\pi}{L} \right) \right) \right) - \frac{a}{L} \times -\frac{1}{k - \frac{2\pi}{L}} \left(\cos \left(x \left(k - \frac{2\pi}{L} \right) \right) \right) \right) \Bigg|_0^L \\ I_2 &= \left(\frac{a}{kL + 2\pi} \times \cos(kL + 2\pi) + \frac{a}{kL - 2\pi} \times \cos(kL - 2\pi) \right) - \left(\frac{a}{kL + 2\pi} \times \cos 0 + \frac{a}{kL - 2\pi} \times \cos 0 \right) \end{aligned}$$

Thus, finally I got:

$$I_2 = \sum_{k=1}^7 \left(\left(\frac{a}{kL + 2\pi} \times \cos(kL + 2\pi) + \frac{a}{kL - 2\pi} \times \cos(kL - 2\pi) \right) - \left(\frac{a}{kL + 2\pi} + \frac{a}{kL - 2\pi} \right) \right) \quad \text{Eq. 7}$$

Now subtracting equation Eq. 7 from Eq. 6 to get the final integral:

$$\begin{aligned} &= \sum_{k=1}^7 \left[\frac{a}{L-k} \times -\frac{1}{2k} (1 - \cos 2L) \right] - \left[\left(\frac{a}{kL + 2\pi} \times \cos(kL + 2\pi) + \right. \right. \\ &\quad \left. \left. \frac{a}{kL - 2\pi} \times \cos(kL - 2\pi) \right) - \left(\frac{a}{kL + 2\pi} + \frac{a}{kL - 2\pi} \right) \right] \quad k \in \mathbb{Z}^+ \end{aligned}$$

Finally, I got the general equation for the A_0 term:

$$A_0 = \sum_{k=1}^7 \frac{-a}{2Lk^2} + \frac{a \cos 2L}{2Lk^2} - \frac{a \cos(kL + 2\pi)}{kL + 2\pi} - \frac{a \cos(kL - 2\pi)}{kL - 2\pi} \quad k \in \mathbb{Z}^+ \quad \text{Eq. 8}$$

Similarly, to compute A_n :

$$\begin{aligned} A_n &= \frac{1}{L} \int_0^L f(x) \times w(x) \times \cos \frac{2\pi nx}{L} dx \\ &= \frac{1}{L} \int_0^L a \left(\sin 2x + \frac{\sin 4x}{2} + \frac{\sin 6x}{3} + \frac{\sin 8x}{4} + \frac{\sin 10x}{5} + \frac{\sin 12x}{6} + \frac{\sin 14x}{7} \right) \\ &\quad \times \left(1 - \cos \frac{2\pi x}{L} \right) \times \cos \frac{2\pi nx}{L} dx \end{aligned}$$

Generalising for k :

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \sin 2kx \times \left(1 - \cos \frac{2\pi x}{L}\right) \times \cos \frac{2\pi nx}{L} dx \quad k, n \in \mathbb{Z}^+$$

Where n denotes the number of the desired harmonic:

Simplifying:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \sin 2kx \times \left(\cos \frac{2\pi nx}{L} - \cos \frac{2\pi x}{L} \times \cos \frac{2\pi nx}{L}\right) dx \quad k, n \in \mathbb{Z}^+$$

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \sin 2kx \times \cos \frac{2\pi nx}{L} - \sin 2kx \times \cos \frac{2\pi x}{L} \times \cos \frac{2\pi nx}{L} dx$$

Now, I could have utilized traditional trigonometric identities similar to the A_0 term, but since this was a more complicated formula, I found a perfect opportunity to use the Euler's formula and complex analysis which I had come across while researching. This approach provides an easier way by making the integral cleaner, by essentially rendering the trigonometric functions into complex exponentials (*Chapter 2 Complex Analysis*), which in turn makes this integration cleaner and closer to real-world signal processing techniques. However, this technique is done by computers because it is a longer path than trigonometric identities, which can be considered a limitation to this method.

Euler's formula states that $e^{i\theta} = \cos\theta + i \sin\theta$ (*Chapter 2 Complex Analysis*) and the general term for representing sine and cosine terms as complex is

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

Substituting these sine and cosine terms in the equation as:

$$\sin 2kx = \frac{e^{i2kx} - e^{-i2kx}}{2i} \quad \cos \frac{2\pi nx}{L} = \frac{e^{i\frac{2\pi nx}{L}} + e^{-i\frac{2\pi nx}{L}}}{2} \quad \cos \frac{2\pi x}{L} = \frac{e^{i\frac{2\pi x}{L}} + e^{-i\frac{2\pi x}{L}}}{2}$$

And now substituting these terms into the equation:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \left(\frac{e^{i2kx} - e^{-i2kx}}{2i} \times \frac{e^{i\frac{2\pi nx}{L}} + e^{-i\frac{2\pi nx}{L}}}{2} \right) dx - \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \left(\frac{e^{i2kx} - e^{-i2kx}}{2i} \times \frac{e^{i\frac{2\pi x}{L}} + e^{-i\frac{2\pi x}{L}}}{2} \times \frac{e^{i\frac{2\pi nx}{L}} + e^{-i\frac{2\pi nx}{L}}}{2} \right) dx$$

Now, to solve for the first term, simplifying it I get:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \left(\frac{e^{i2x(k+\frac{\pi n}{L})} + e^{i2x(k-\frac{\pi n}{L})} - e^{-i2x(k-\frac{\pi n}{L})} - e^{-i2x(k+\frac{\pi n}{L})}}{4i} \right)$$

Integrating this simplified version, I get:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k \cdot 4i} \left[\frac{e^{i2L(k+\frac{\pi n}{L})} - 1}{i2(k+\frac{\pi n}{L})} + \frac{e^{i2L(k-\frac{\pi n}{L})} - 1}{i2(k-\frac{\pi n}{L})} + \frac{e^{-i2L(k-\frac{\pi n}{L})} - 1}{-i2(k-\frac{\pi n}{L})} + \frac{e^{-i2L(k+\frac{\pi n}{L})} - 1}{-i2(k+\frac{\pi n}{L})} \right] \quad \text{Eq. 9}$$

Now simplifying the second term:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \frac{e^{i2kx} - e^{-i2kx}}{2i} \times \frac{\left(e^{\frac{i2\pi x}{L}(1+n)} + e^{\frac{i2\pi x}{L}(1-n)} + e^{\frac{i2\pi x}{L}(n-1)} + e^{\frac{i2\pi x}{L}(-1-n)} \right)}{4}$$

$$= \sum_{k=1}^7 \frac{a}{L \cdot k \cdot 8i} \int_0^L \left[\left(e^{\frac{i2\pi x}{L}(1+n)+i2kx} + e^{\frac{i2\pi x}{L}(1-n)+i2kx} + e^{\frac{i2\pi x}{L}(n-1)+i2kx} + e^{\frac{i2\pi x}{L}(-1-n)+i2kx} \right) - \left(e^{\frac{i2\pi x}{L}(1+n)-i2kx} + e^{\frac{i2\pi x}{L}(1-n)-i2kx} + e^{\frac{i2\pi x}{L}(n-1)-i2kx} + e^{\frac{i2\pi x}{L}(-1-n)-i2kx} \right) \right]$$

And integrating this part, I get:

$$= \sum_{k=1}^7 \frac{a}{L \cdot k \cdot 8i} \left[\left(\frac{e^{i2\pi(1+n)+i2kL} - 1}{i2\pi(1+n)+i2kL} + \frac{e^{i2\pi(1-n)+i2kL} - 1}{i2\pi(1-n)+i2kL} + \frac{e^{i2\pi(n-1)+i2kL} - 1}{i2\pi(n-1)+i2kL} + \frac{e^{i2\pi(-1-n)+i2kL} - 1}{i2\pi(-1-n)+i2kL} \right) - \left(\frac{e^{i2\pi(1+n)-i2kL} - 1}{i2\pi(1+n)-i2kL} + \frac{e^{i2\pi(1-n)-i2kL} - 1}{i2\pi(1-n)-i2kL} + \frac{e^{i2\pi(n-1)-i2kL} - 1}{i2\pi(n-1)-i2kL} + \frac{e^{i2\pi(-1-n)-i2kL} - 1}{i2\pi(-1-n)-i2kL} \right) \right] \quad \text{Eq. 10}$$

Now, finally putting together Eq. 9 and Eq. 10 back together to get the A_n term:

$$A_n = \sum_{k=1}^7 \left[\frac{a}{L \cdot k \cdot 4i} \left\{ \frac{e^{i2L(k+\frac{\pi n}{L})} - 1}{i2(k+\frac{\pi n}{L})} + \frac{e^{i2L(k-\frac{\pi n}{L})} - 1}{i2(k-\frac{\pi n}{L})} + \frac{e^{-i2L(k-\frac{\pi n}{L})} - 1}{-i2(k-\frac{\pi n}{L})} + \frac{e^{-i2L(k+\frac{\pi n}{L})} - 1}{-i2(k+\frac{\pi n}{L})} \right\} - \frac{a}{L \cdot k \cdot 8i} \left\{ \left(\frac{e^{i2\pi(1+n)+i2kL} - 1}{i2\pi(1+n)+i2kL} + \frac{e^{i2\pi(1-n)+i2kL} - 1}{i2\pi(1-n)+i2kL} + \frac{e^{i2\pi(n-1)+i2kL} - 1}{i2\pi(n-1)+i2kL} + \frac{e^{i2\pi(-1-n)+i2kL} - 1}{i2\pi(-1-n)+i2kL} \right) - \left(\frac{e^{i2\pi(1+n)-i2kL} - 1}{i2\pi(1+n)-i2kL} + \frac{e^{i2\pi(1-n)-i2kL} - 1}{i2\pi(1-n)-i2kL} + \frac{e^{i2\pi(n-1)-i2kL} - 1}{i2\pi(n-1)-i2kL} + \frac{e^{i2\pi(-1-n)-i2kL} - 1}{i2\pi(-1-n)-i2kL} \right) \right\} \right] \quad k, n \in \mathbb{Z}^+ \quad \text{Eq. 11}$$

Similarly, the B_n term, which was:

$$B_n = \frac{1}{L} \int_0^L f(x) \times w(x) \times \sin \frac{2\pi nx}{L} dx$$

Was calculated to be:

$$B_n = \sum_{k=1}^7 \frac{a}{L \cdot k} \int_0^L \sin 2kx \times \left(1 - \cos \frac{2\pi x}{L} \right) \times \sin \frac{2\pi nx}{L} dx$$

Which, similarly, can be integrated using complex analysis and Euler's formula to get:

$$B_n = \sum_{k=1}^7 \left[\frac{-a}{L \cdot k \cdot 4} \left\{ \frac{e^{i2L(k+\frac{\pi n}{L})}-1}{i2(k+\frac{\pi n}{L})} - \frac{e^{i2L(k-\frac{\pi n}{L})}-1}{i2(k-\frac{\pi n}{L})} - \frac{e^{i2L(-k+\frac{\pi n}{L})}-1}{i2(-k+\frac{\pi n}{L})} + \frac{e^{i2L(-k-\frac{\pi n}{L})}-1}{i2(-k-\frac{\pi n}{L})} \right\} + \right. \\ \left. \frac{a}{L \cdot k \cdot 8} \left\{ \left(\frac{e^{i2\pi(1+n)+i2kL}-1}{i2\pi(1+n)+i2kL} - \frac{e^{i2\pi(1-n)+i2kL}-1}{i2\pi(1-n)+i2kL} + \frac{e^{i2\pi(n-1)+i2kL}-1}{i2\pi(n-1)+i2kL} - \frac{e^{i2\pi(-1-n)+i2kL}-1}{i2\pi(-1-n)+i2kL} \right) - \right. \right. \\ \left. \left. \left(\frac{e^{i2\pi(1+n)-i2kL}-1}{i2\pi(1+n)-i2kL} - \frac{e^{i2\pi(1-n)-i2kL}-1}{i2\pi(1-n)-i2kL} + \frac{e^{i2\pi(n-1)-i2kL}-1}{i2\pi(n-1)-i2kL} - \frac{e^{i2\pi(-1-n)-i2kL}-1}{i2\pi(-1-n)-i2kL} \right) \right\} \right] \quad k, n \in \mathbb{Z}^+ \quad \text{Eq. 12}$$

Where, for both Eq. 11 and Eq. 12:

- a – Amplitude of negative ramp sawtooth wave
- k – Index of summation of the sawtooth wave
- L – Total time interval of the sample of audio being analysed
- n – Number of desired harmonic

Using Wolfram Alpha's calculator (Wolfram Alpha LLC), I calculated the coefficients for the high E string using Eq. 8, Eq. 11 and Eq. 12. I got the A_0 term as **0.0450565864**. This means that the recorded wave oscillates almost symmetrically around zero, with a slight positive offset above the X-axis.

(I used 10 decimal places since all answers are very small) Next, I calculated the A_n and B_n coefficient's value for the first 20 harmonics as defined by the notation 'n'. I chose 20 since it provides a good balance between capturing the key harmonic structure of the guitar strings and keeping the calculations manageable, as higher harmonics contribute minimally to the overall sound. I substituted $L = 0.022$, $a = 0.315$, and $n = 1$ to 20 in Eq. 8, Eq. 11 and Eq. 12 to get the following results:

Harmonic (n)	A_n term	B_n term	Harmonic (n)	A_n term	B_n term
1	-0.015755816	0.021095581	11	0.0012059	4.46442×10^{-5}
2	-0.000770476	0.01342823	12	0.000981867	2.64836×10^{-5}
3	0.000378253	0.006922992	13	0.000817466	1.52698×10^{-5}
4	-0.000546204	0.007439327	14	0.000692547	8.00036×10^{-6}

Harmonic (n)	A_n term	B_n term	Harmonic (n)	A_n term	B_n term
5	0.002020522	0.00614315	15	0.000595042	3.1227×10^{-6}
6	0.005424407	0.004224352	16	0.000517276	-2.32083×10^{-7}
7	0.004830522	0.000897514	17	0.000454144	-2.58×10^{-6}
8	0.002868167	0.000286289	18	0.000402121	-4.24203×10^{-6}
9	0.002017289	0.000137862	19	0.0003587	-5.42544×10^{-6}
10	0.001526092	7.62396×10^{-5}	20	0.000322057	-6.26819×10^{-6}

Table 2: Coefficients of the first 20 harmonics

Now, I have the amplitudes of the sine and cosine terms present in the Fourier series of the wave up to 20 harmonics. Now that I have those amplitudes of the sine and cosine terms for the Fourier series of my wave up through 20 harmonics, what I want is to know for each harmonic how much it will contribute to a total wave--essentially what the 'strength' of every harmonic is together. I can get this magnitude C_n using the following formula (A. Peacock):

$$\sqrt{A_n^2 + B_n^2} = C_n$$

Calculating this in Microsoft Excel 2021, I obtained the following values:

Harmonic (n)	C_n term	Harmonic (n)	C_n term
1	0.026330007	11	0.001206726
2	0.013450316	12	0.000982224
3	0.006933317	13	0.000817608
4	0.007459351	14	0.000692593
5	0.006466901	15	0.00059505
6	0.00687527	16	0.000517276
7	0.004913194	17	0.000454152
8	0.00288242	18	0.000402143
9	0.002021994	19	0.000358741
10	0.001527995	20	0.000322118

Table 3: C_n term values

Thus, I saw that the maximum C_n value was of the first harmonic itself, meaning it had was the most dominant frequency among all others. And since the frequency descended as the harmonics increases, I can safely assume that there are no other frequencies more dominant than the 1st or fundamental frequency in this case. Now, to find the actual frequency from the harmonic number, I used the number of the harmonic and multiplied it with the fundamental frequency:

$$n \times \text{fundamental frequency} = \text{desired frequency}$$

The fundamental frequency is as follows:

$$\text{fundamental frequency} = \frac{1}{\text{Time period of 1 oscillation}}$$

Since the in the interval of 0.022 s there were seven oscillations, hence the fundamental frequency is:

$$\frac{1}{\frac{0.022}{7}} = \frac{1}{\frac{22}{7 \times 1000}} = \frac{7 \times 1000}{22} = \mathbf{318.18 \text{ Hz}}$$

Hence, the dominant frequency for the high E string is 318.18 Hz. To check the accuracy of the result of my Fourier analysis, I used the ‘Plot Spectrum’ function in Audacity, which itself performs Fourier analysis to perform frequency analysis, and I can choose a windowing function too, as shown in Figure 5 below:

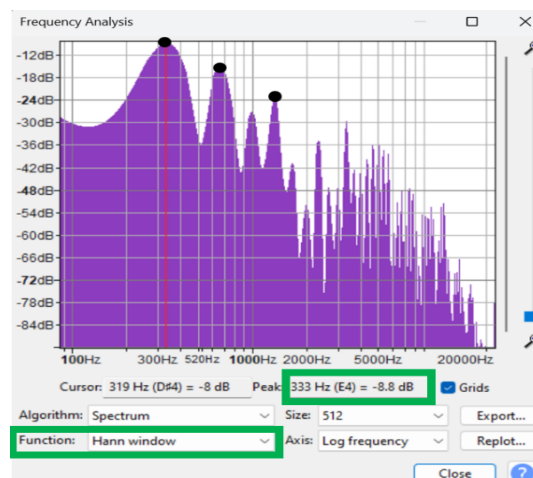


Figure 5: Frequency spectrum of the high E string

The plot shows the highest peak to be of the frequency 333 Hz using the Hann window, which aligns with my exploration. Thus, my results have a small error as compared to this computed value, which comes to **4.45%**, and also align closely to the literature value of the frequency, 329.63 Hz, with error **4.51%**, both are shown in Eq. 13 and Eq. 14. The causes of these error have been discussed in the Limitations section later.

$$= \frac{(333-318)}{333} \times 100 = 4.45\% \quad \text{Eq. 13}$$

$$= \frac{(329.63-318)}{329.63} \times 100 = 4.51\% \quad \text{Eq. 14}$$

In Figure 5, I have also marked 3 black dots, which show that the top 3 dominant frequencies are the 1st, 2nd and 4th in descending order. These values correspond with the descending values of the C_n term in Table 3, which show the values 0.026330007, 0.013450316, and 0.007459351 of the 1st, 2nd and 4th harmonic respectively, thus highlighting the accuracy of this exploration. However, a limitation here is that the spectrum shows the frequencies with its relative amplitude in the sound sample, and does not provide exact magnitudes the way I computed the C_n terms. Furthermore, the relative amplitude of higher harmonics in Figure 5 can be seen to rise slightly and then fall again, which does not align with the assumption I made that the magnitudes of higher harmonic frequencies show a decreasing trend. Thus, I cannot be completely sure that the frequencies showcased in the spectrum align perfectly with my results.

Low E string: Calculations and Analysis

Next, I performed this whole process with the recording of the low E string. The curve that was recorded is shown in Figure 1. I again chose a part with 7 oscillations, because using the same number of oscillations ensures that my analysis captures a complete and comparable harmonic pattern across both samples, regardless of their individual time periods. This time the

time period of the audio sample selected was from 1.247 to 1.332 s, which makes $L = 0.085$ s.

The modelled function fitted on the curve and its equation is:

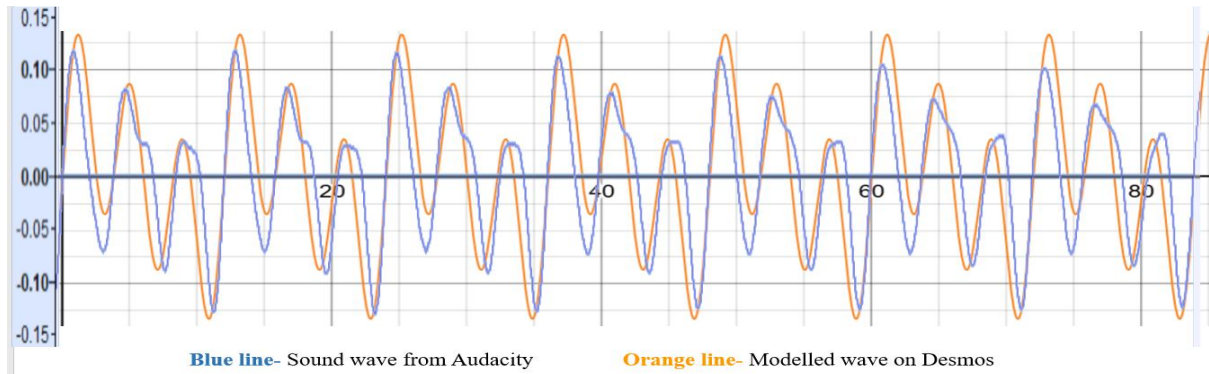


Figure 6: Matching modelled function to audio sample of low E string

$$g(x) = 0.05 \left(\sin \frac{2800\pi x}{17} + \frac{\sin \frac{5600\pi x}{17}}{2} + \frac{\sin \frac{8400\pi x}{17}}{2} \cdot 5 \right) \quad \text{Eq. 15}$$

The Hann windowed function was:

$$g_{Hann}(x) = 0.05 \left(\sin \frac{2800\pi x}{17} + \frac{\sin \frac{5600\pi x}{17}}{2} + \frac{\sin \frac{8400\pi x}{17}}{2} \times 5 \right) \times 0.5 \times \left(1 - \cos \frac{2\pi x}{0.085} \right) \quad \text{Eq. 16}$$

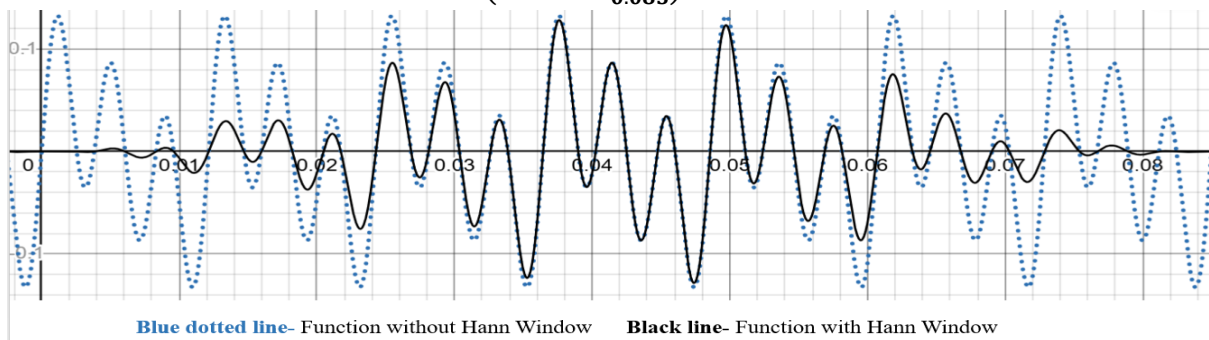


Figure 7: Comparing function with and without Hann window

Once again, using Eq. 8, Eq. 11 and Eq. 12, however this time plugging in $L = 0.085$, $a = 0.05$ and $n = 1$ to 20, and calculating the C_n similar to the high E string, I got the following values.

Harmonic (n)	A_n term	B_n term	C_n term
1	-4.53822×10^{-16}	1.19581×10^{-19}	4.53822×10^{-16}
2	-1.1604×10^{-16}	2.64075×10^{-19}	1.1604×10^{-16}

Harmonic (n)	A_n term	B_n term	C_n term
3	1.71018×10^{-16}	0.025	0.025
4	2.49253×10^{-15}	-0.0208333	0.0208333
5	8.2669×10^{-17}	-4.44443×10^{-18}	8.27884×10^{-17}
6	2.45806×10^{-16}	-0.0125	0.0125
7	-5.39817×10^{-17}	-2.79023×10^{-19}	5.39825×10^{-17}
8	1.51115×10^{-16}	1.18385×10^{-17}	1.51578×10^{-16}
9	3.33764×10^{-15}	-0.00625	0.00625
10	-1.55212×10^{-16}	7.71298×10^{-18}	1.55403×10^{-16}
11	-1.94634×10^{-15}	-5.58045×10^{-18}	1.94635×10^{-15}
12	-4.13679×10^{-17}	-4.94269×10^{-18}	4.16622×10^{-17}
13	-1.88794×10^{-15}	-0.0125	0.0125
14	-4.41422×10^{-15}	-1.97309×10^{-18}	4.41422×10^{-15}
15	-2.32779×10^{-15}	-0.00625	0.00625
16	3.31426×10^{-15}	7.57845×10^{-18}	3.31427×10^{-15}
17	-6.13017×10^{-16}	1.53462×10^{-18}	6.13019×10^{-16}
18	-1.7595×10^{-15}	0.0125	0.012
19	-2.99045×10^{-15}	-2.29795×10^{-17}	3.77139×10^{-17}
20	2.5137×10^{-17}	-3.18883×10^{-19}	2.5139×10^{-17}

Table 4: Coefficients of Fourier analysis of low E string

These values show that the dominant frequency here is the 3rd harmonic, followed by the 4th and 1st harmonic. Here, the fundamental frequency was calculated to be **82.35 Hz**, thus the dominant frequencies, in descending order of nearest whole number frequency, are **247.05**

Hz, 329.40 Hz and 82.35 Hz. This can be verified by the 3 highest frequencies seen in the frequency spectrum generated of this wave:

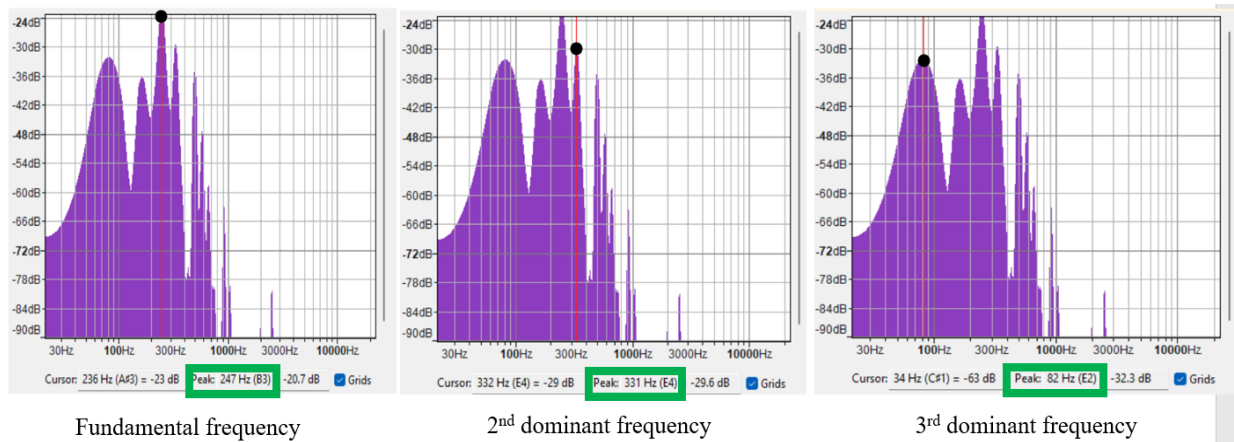


Figure 8: Frequency analysis spectrum by Audacity

Thus, the error calculated for each frequency was **0.02%, 0.48% and 0.43%** respectively, and **0.07% error of the fundamental frequency with the SPN value** (Liu et al.)(Equations used were similar to Eq. 13 and Eq. 14). These very low values of error shows that the modelled equation here is a better approximation of the sound wave than the modelled function of the high E string was for its sound wave.

Another important observation in this experiment is the existence of a frequency around 318.18 Hz, which is the second most dominant frequency in the wave. This frequency is essentially a harmonic of the fundamental frequency of the high E string that is about 330 Hz. This result further confirms that the fundamental vibrational mode of the high E string is actually in the wave. This harmonic reveals that, whenever the low E string is struck, the high E string reacts not only to the fundamental harmonic of the low E string but also to some of its harmonics. This phenomenon is indicative of the coupled harmonic system in the guitar, where vibrations propagate through the instrument's body and are then picked up by the strings themselves. Sympathetic vibration principles in which the nearby strings resonate thereby causing them to vibrate upon the primary vibration of the concerned string are based on this evidence. Mathematical analysis of the problem by decomposition of the Fourier series further

justifies the aim of this exploration: showing harmonic components of the sound wave from the low E string match with the resonance frequency of the high E string.

Conclusion:

The work was a detailed exploration on how to describe the harmonic relations between the high E and low E guitar strings with Fourier series. The output indicates that for the high E, the dominating frequency was at 318.18 Hz with negligible error relative to the Audacity frequency spectrum as well as in comparison with literature values. In the case of the low E, it has been dominated by the third harmonic with an error of 0.07%. Despite the difficulties of white noise and manual wave modelling, the Fourier series method was able to decompose the sound waves and identify the important frequencies. Overall, the exploration successfully fulfilled its aim of analysing the harmonic content of both strings, providing valuable insight into their frequency structures.

Limitations

- Approximating a waveform using a Fourier series, overshoot by Gibbs has not been explored in this experiment. This can be minimized if more harmonics are included, or using other windowing techniques such as the Hamming window or Blackman window.
- Audacity's spectrum plot measures based on energy distribution and not the harmonic contribution, and therefore discrepancies arise with Fourier coefficient analysis.
- Sound waves have nonlinearities or extra parts that may not be included in the Fourier series, so the model is not exact. Plus, the model approximation is also not perfectly aligned with the wave recorded.
- A spectrum plot depends on energy distribution, so the real strength of individual frequencies is hidden. Harmonic contributions appear in Fourier coefficients (C_n).

- Since it is the frequency that generates periodicity, it is the dominant frequency, and higher harmonics may have larger coefficients, requiring deeper analysis to be sure of picking the right harmonic.
- Fourier series assumes complete periodicity over the chosen time interval. Deviation from real wave periods may lead to inaccuracy in the outcome.
- Out-of-band contributions, such as noise or a very loud section of the audio, will affect the analysis and the spectrum plot and therefore can result in errors.
- The Hann window has the least amount of leakage into the spectrum, but at a slight penalty by attenuating the amplitudes near the wave's ends.

Extension of research

- The Least Squares Spectral Analysis (LSSA) could be employed in this investigation. This is a technique to reduce the noise without knowing the magnitudes of the noise constituents. This technique includes fitting a model (such as sinusoidal functions) to observed data by minimizing the sum of squared differences between the observed and predicted values. (Petr Vaníček).
- Another extension could be a nonlinear Fourier transform, possibly as an extension to analyse signals from nonlinear systems, providing another angle of thought on complex or chaotic sound signals where traditional Fourier analysis may be less effective.
- Using FFT would have made the analysis faster and more efficient, especially for handling non-periodic components. However, it would have reduced the clarity of individual harmonic contributions, which were better interpreted through Fourier coefficients in my approach (“Difference between Fourier Series and Fourier Transform”).

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